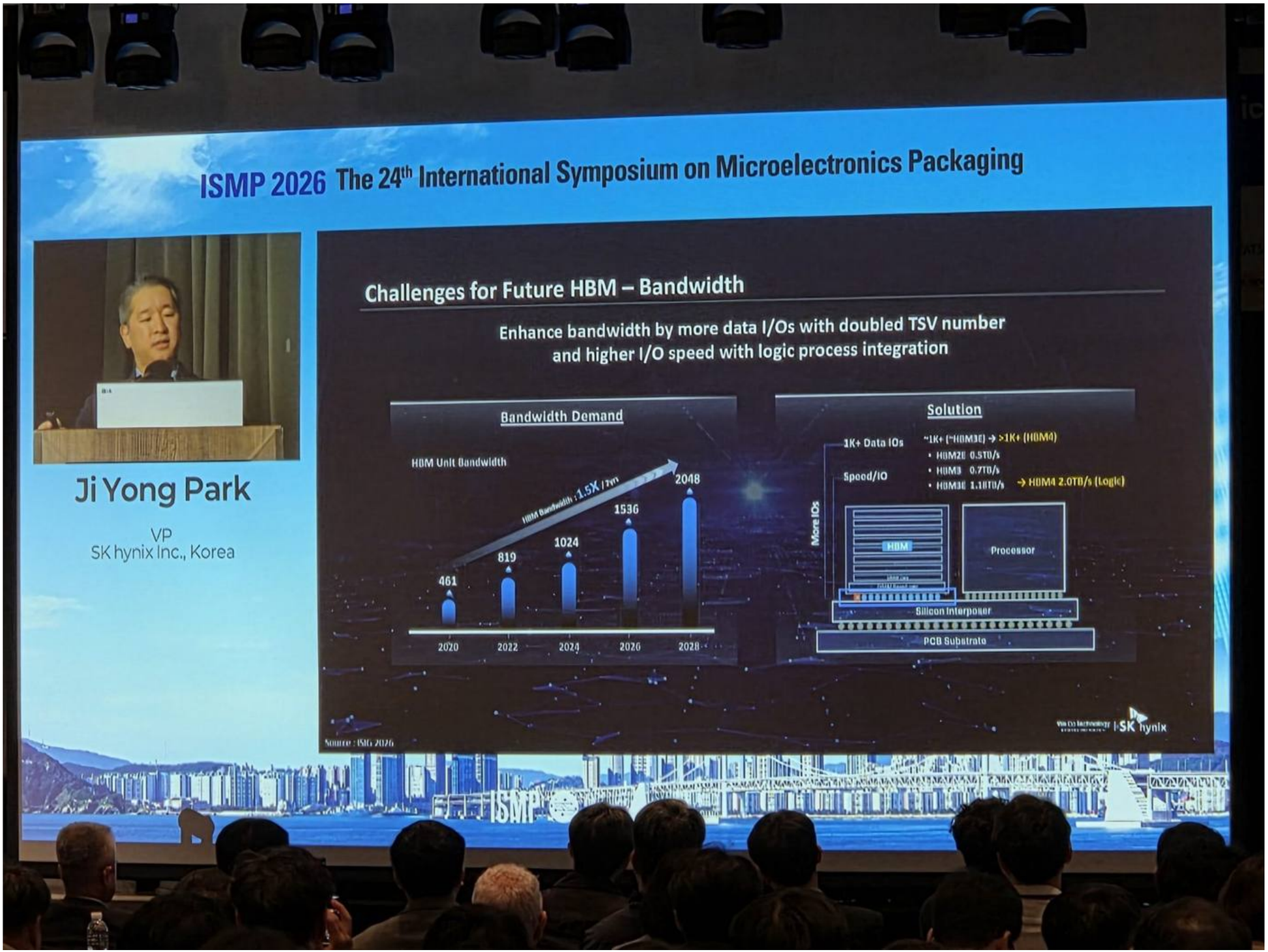
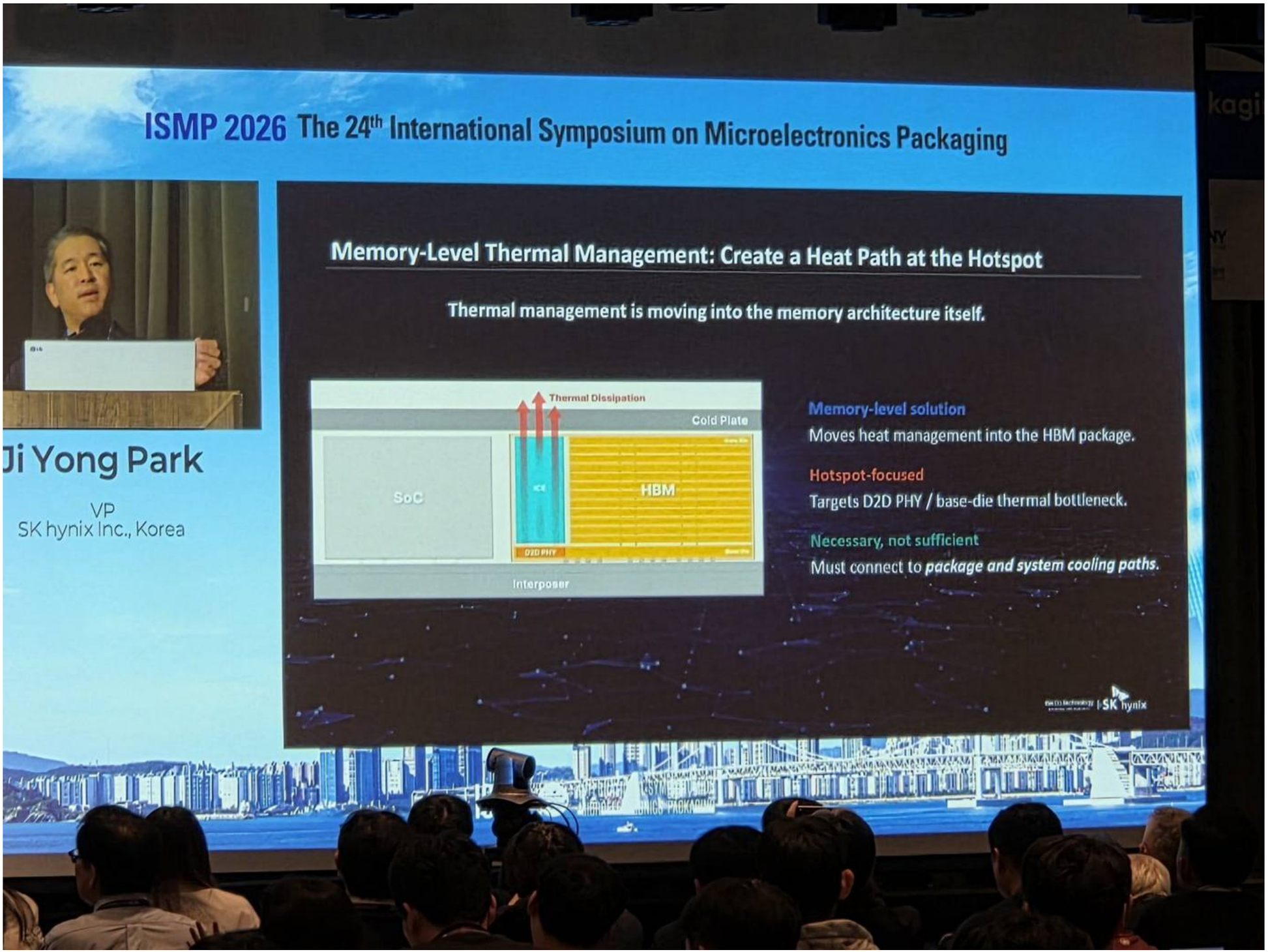
T17 AI 메모리의 집적 냉각과 수직 확장

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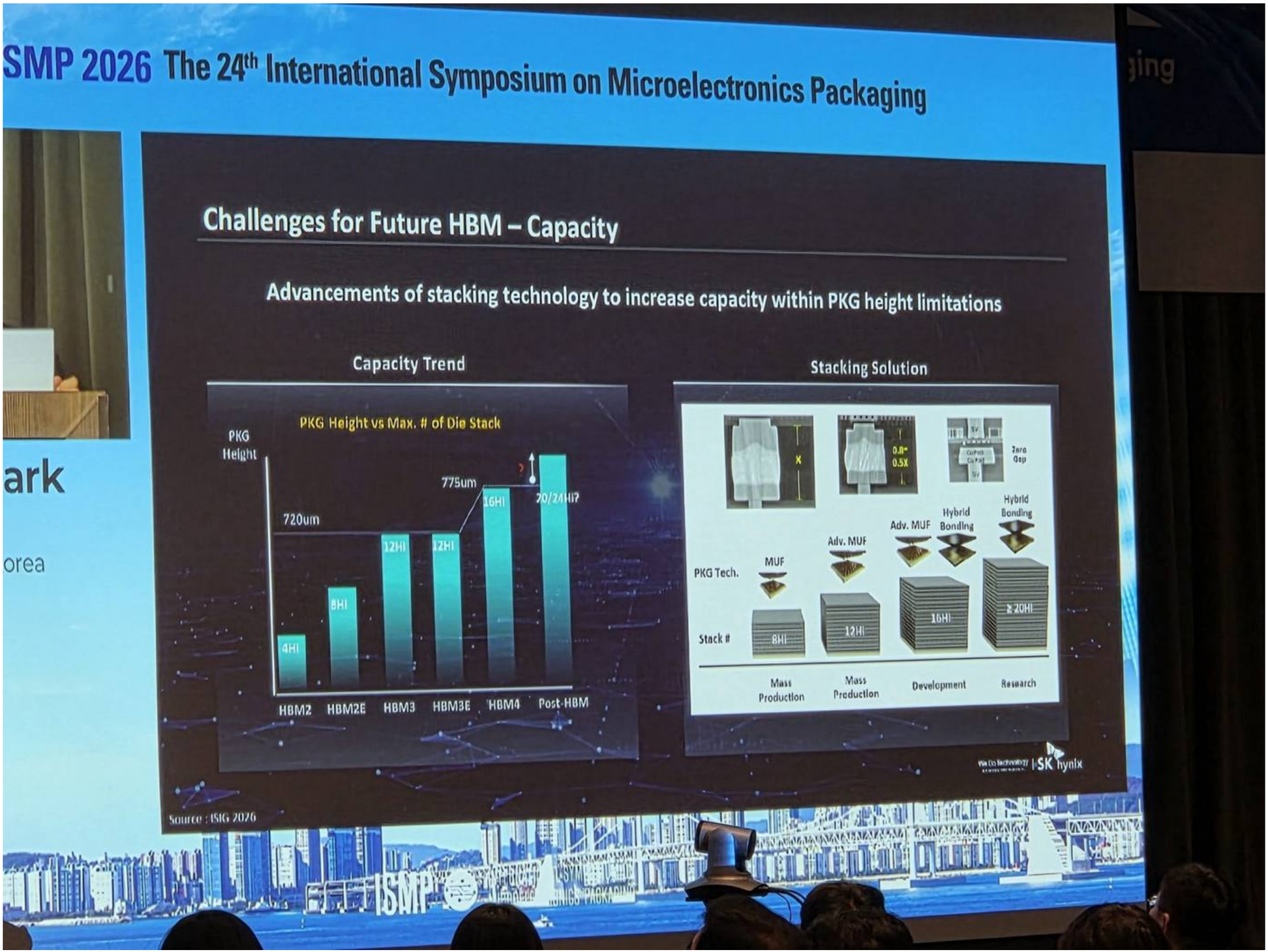
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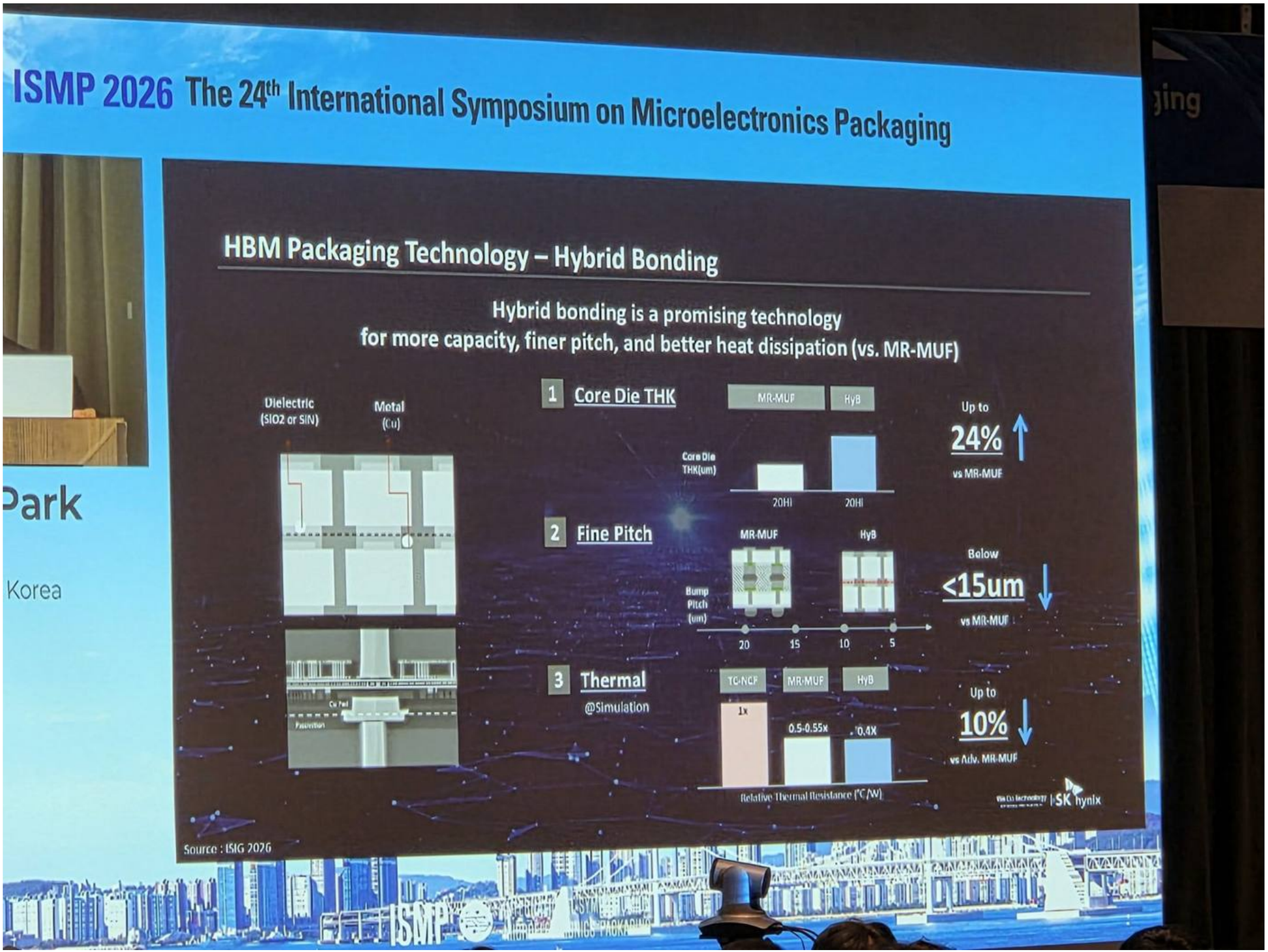
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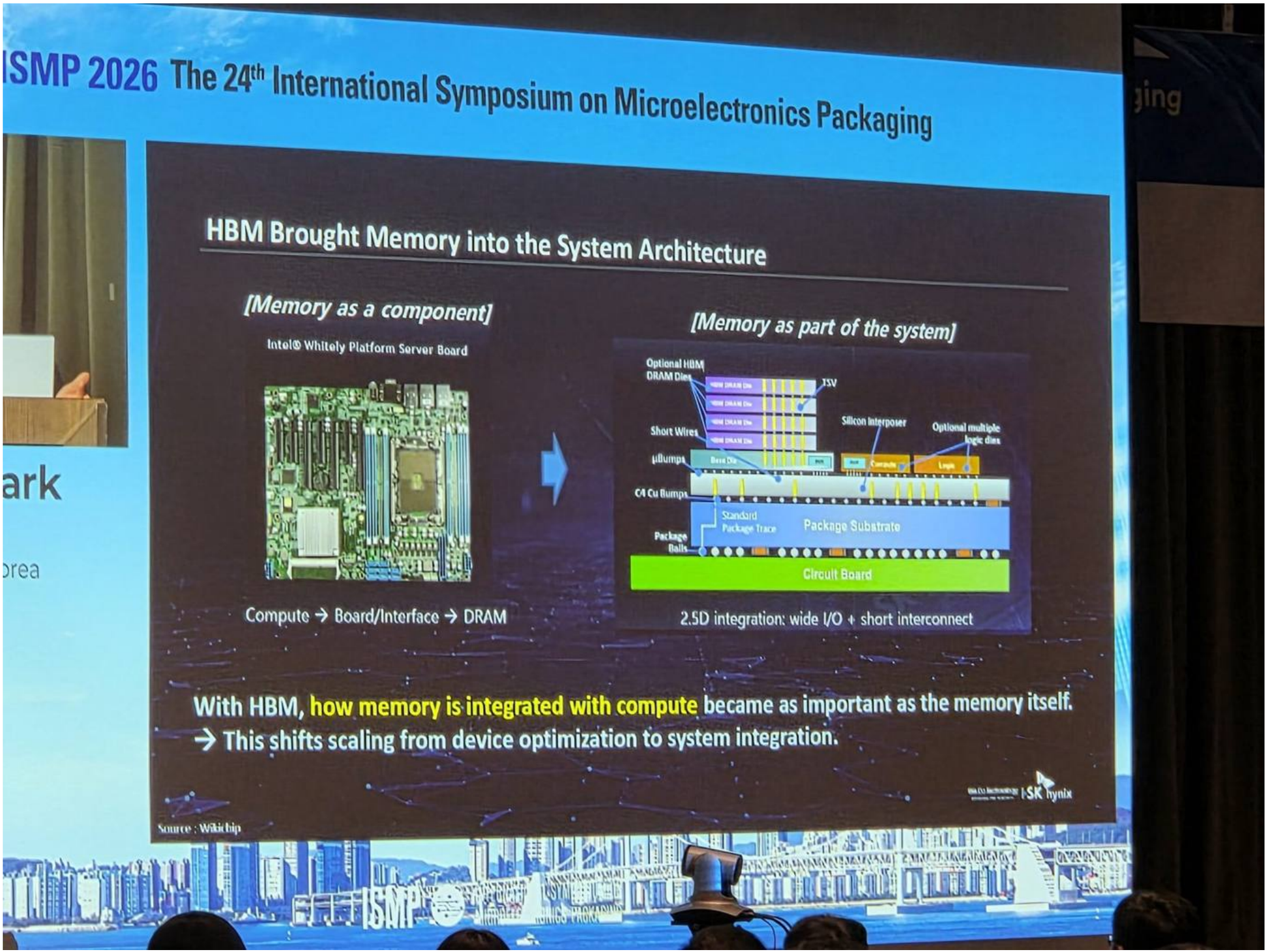
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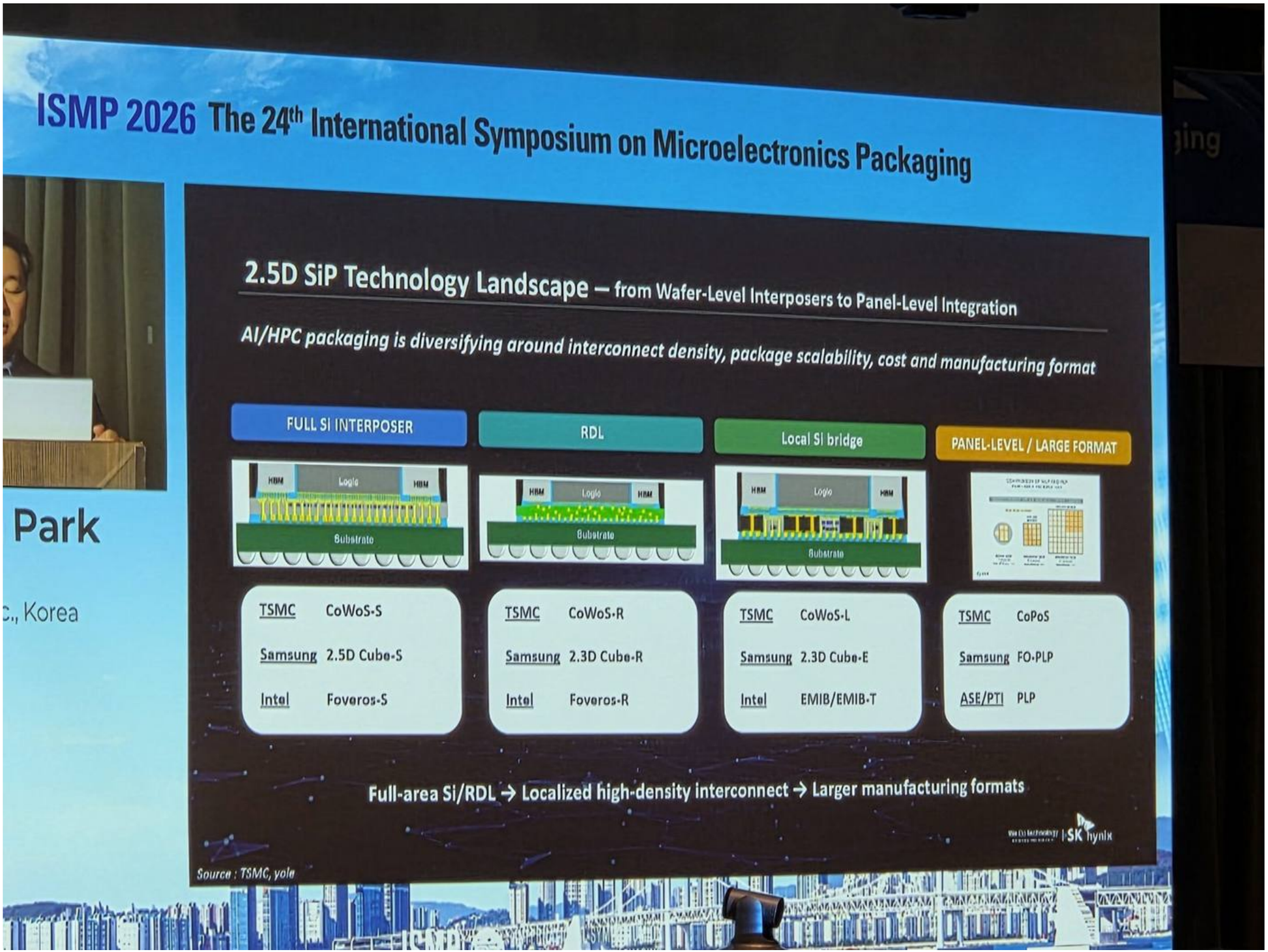
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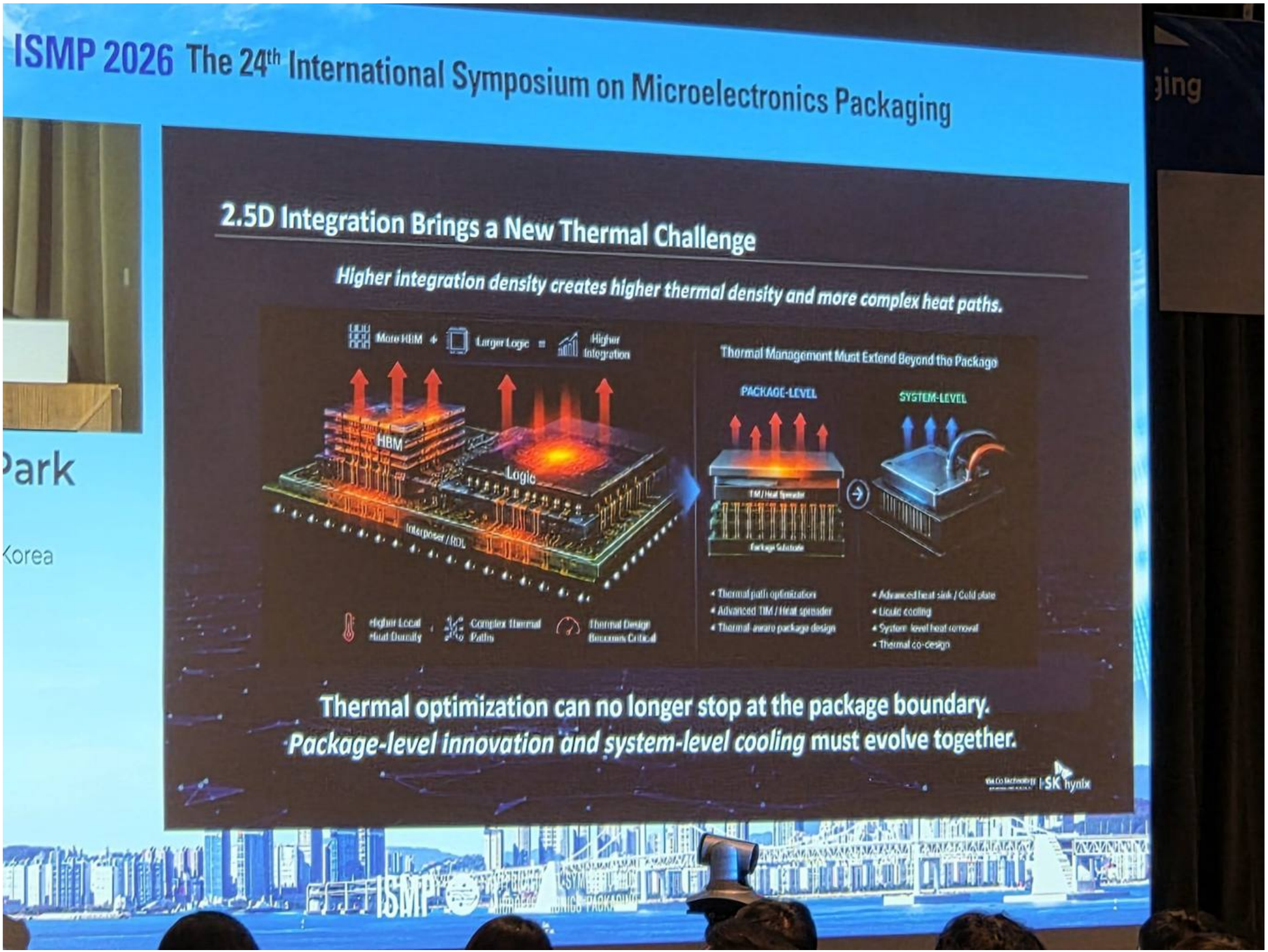
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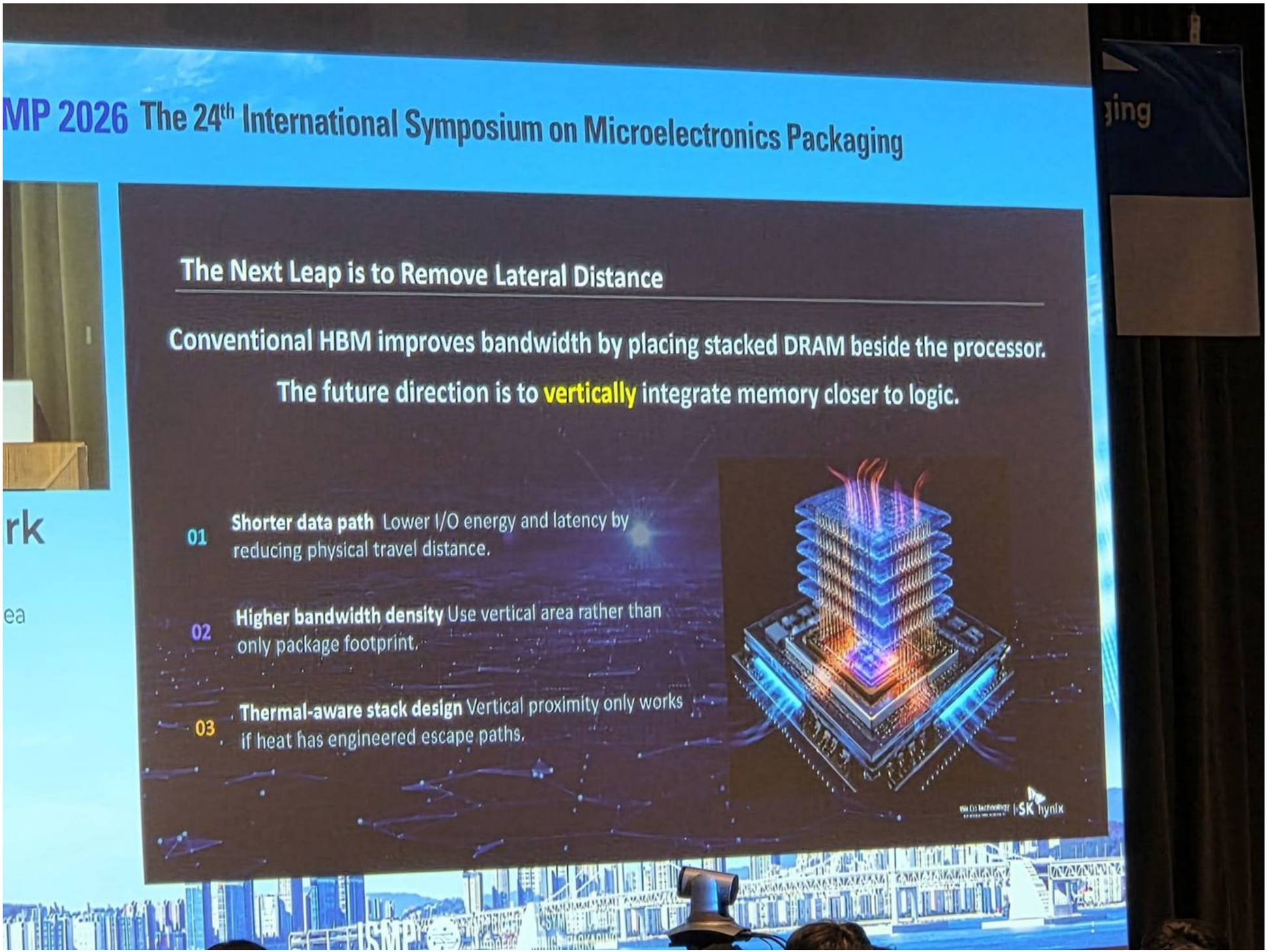
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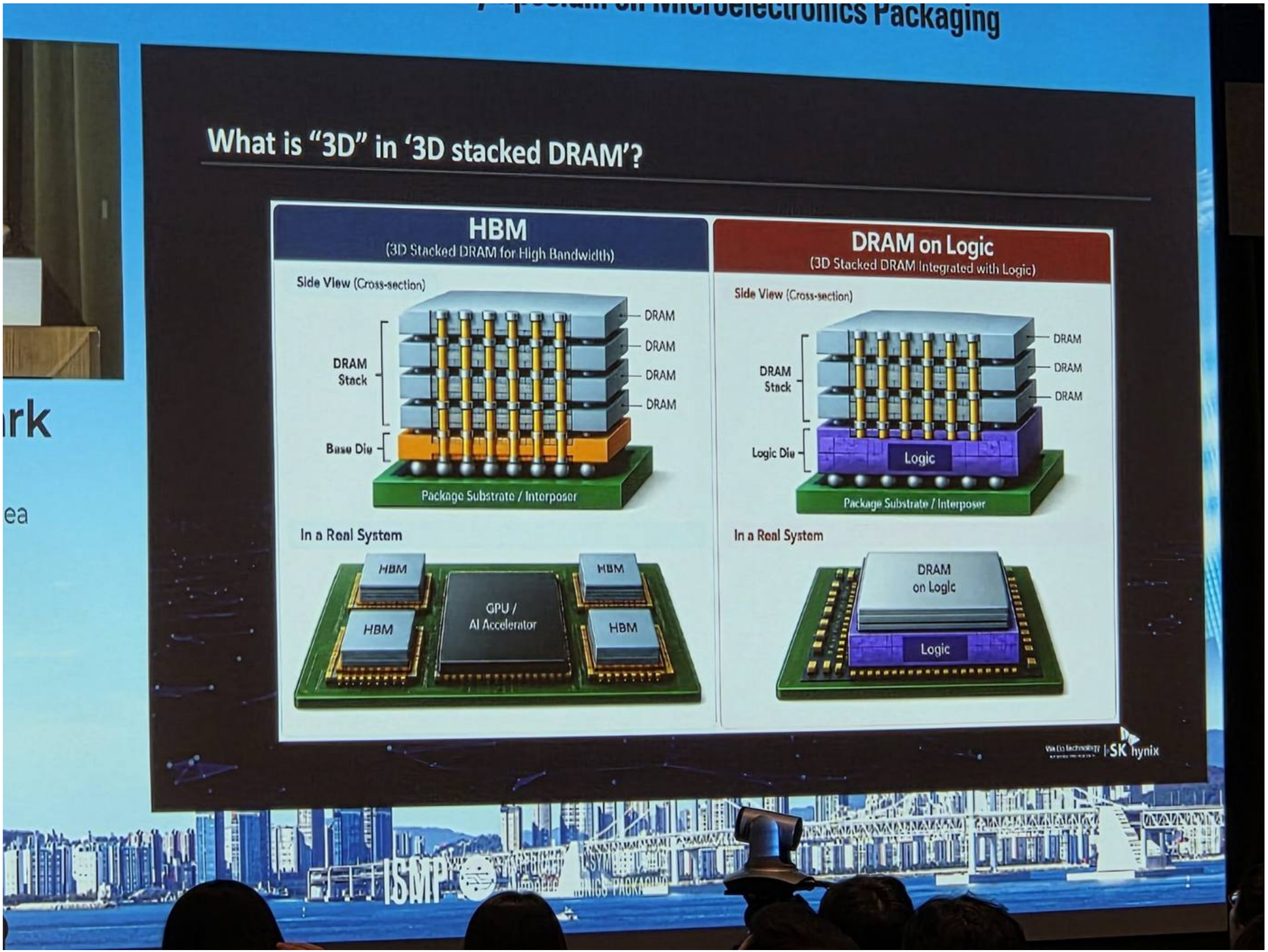
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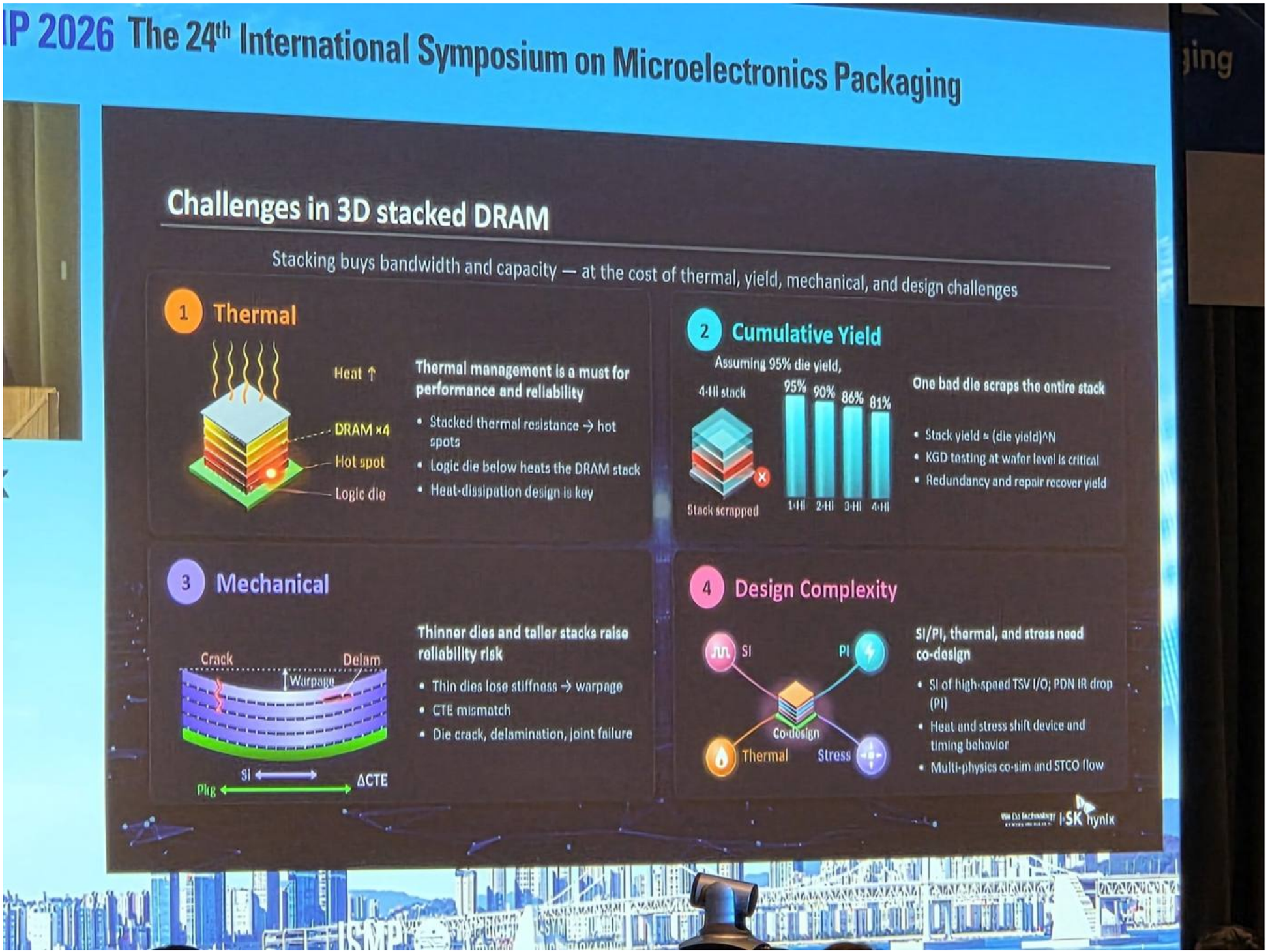
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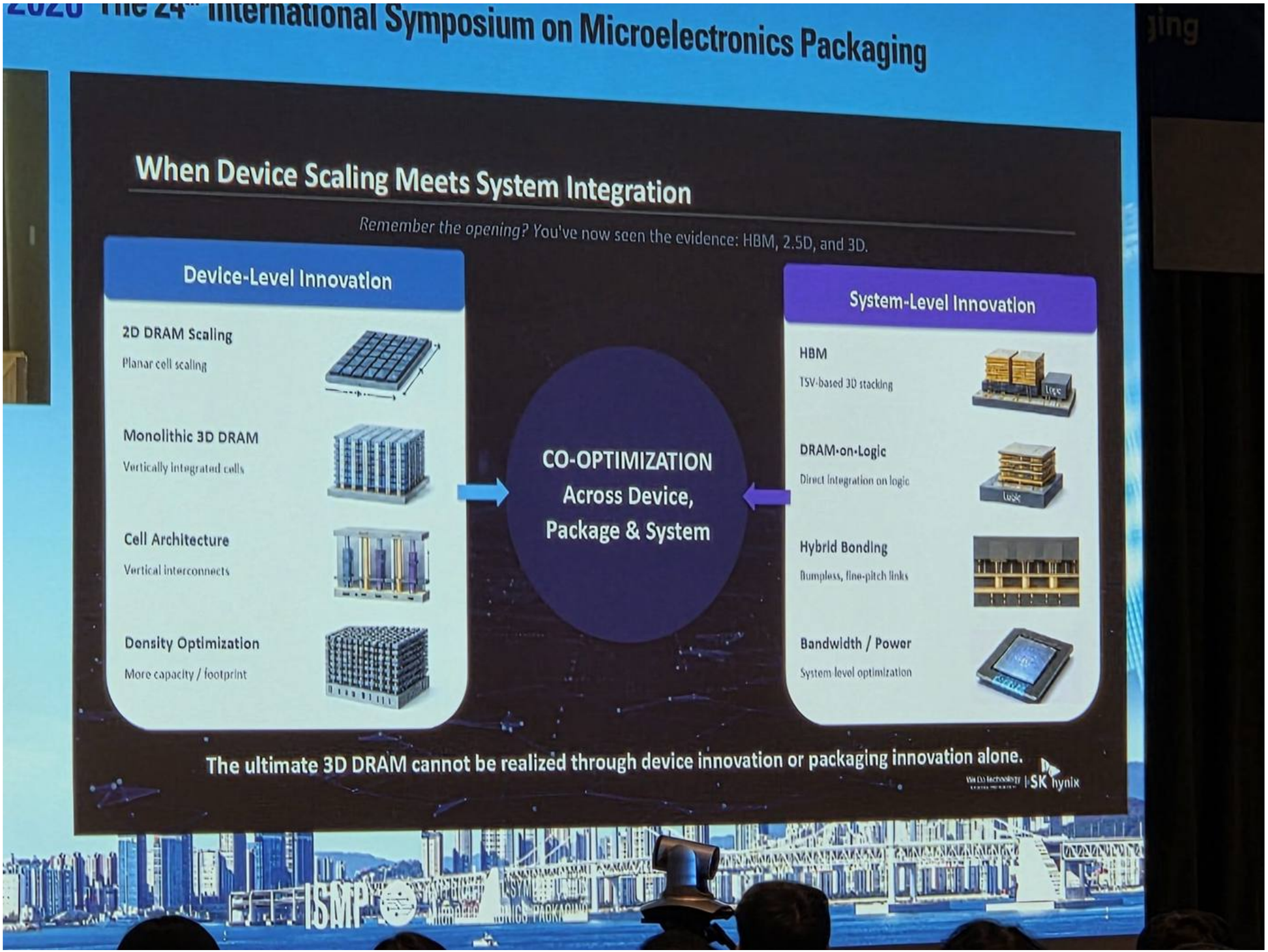
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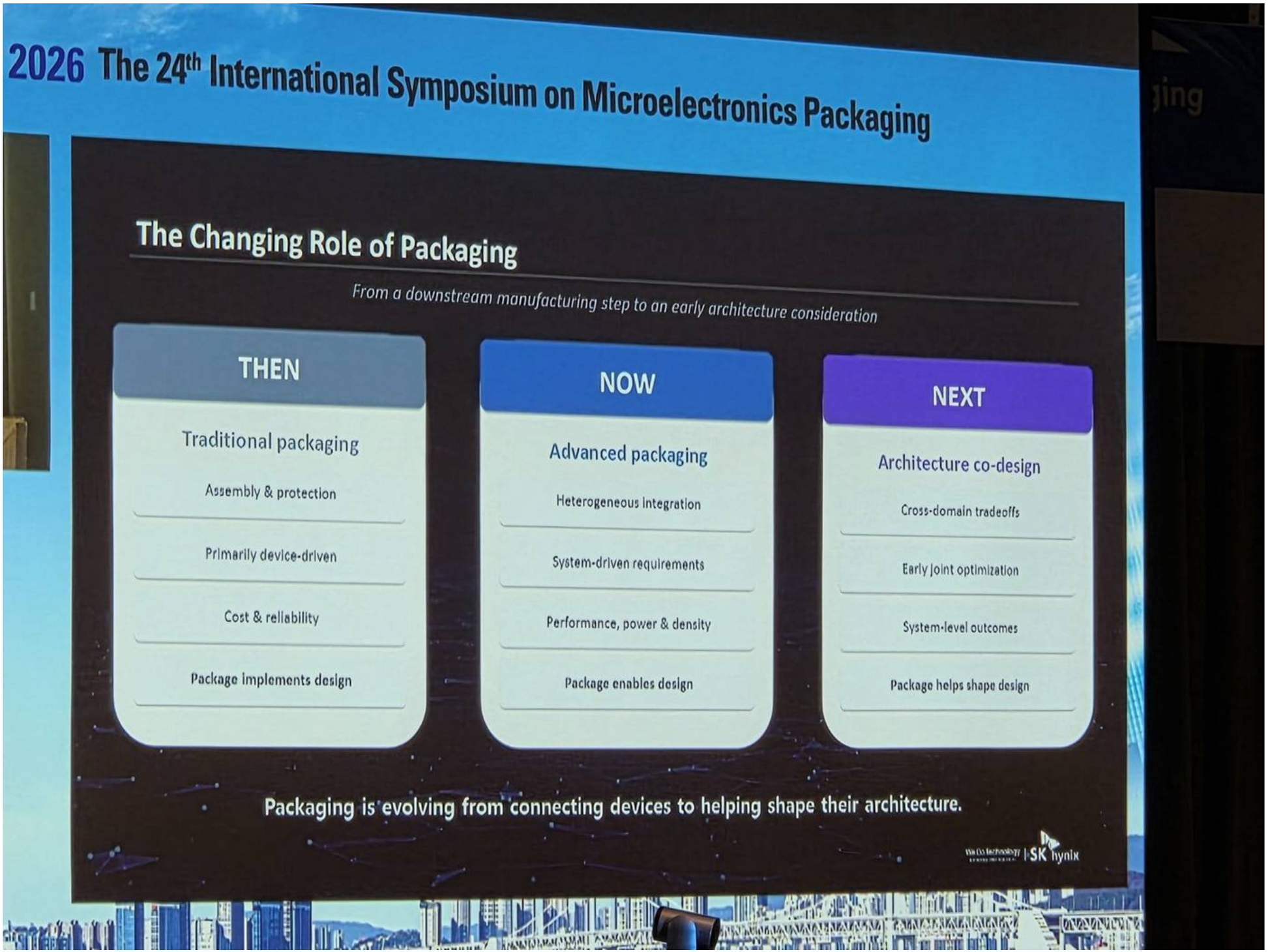
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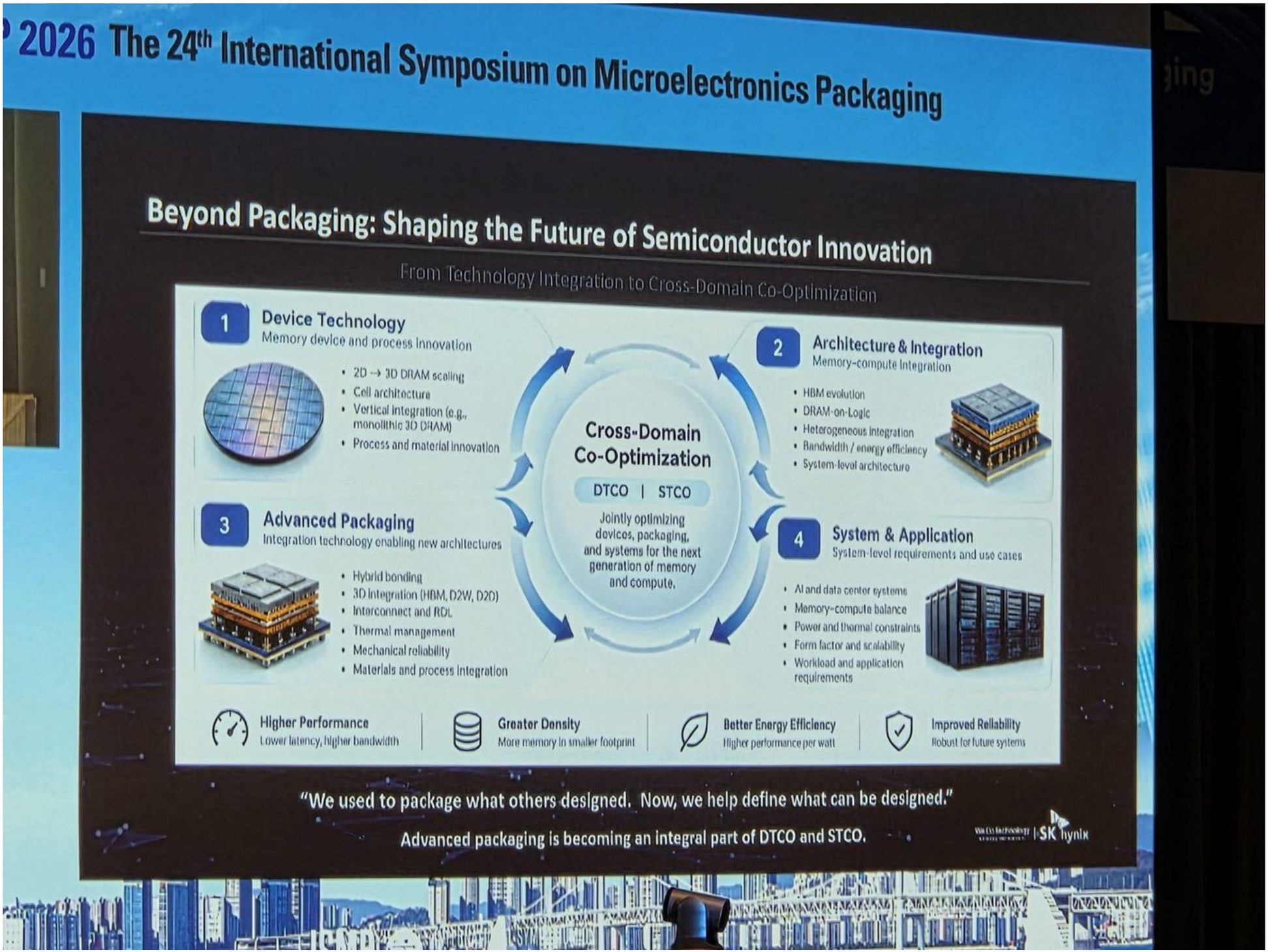
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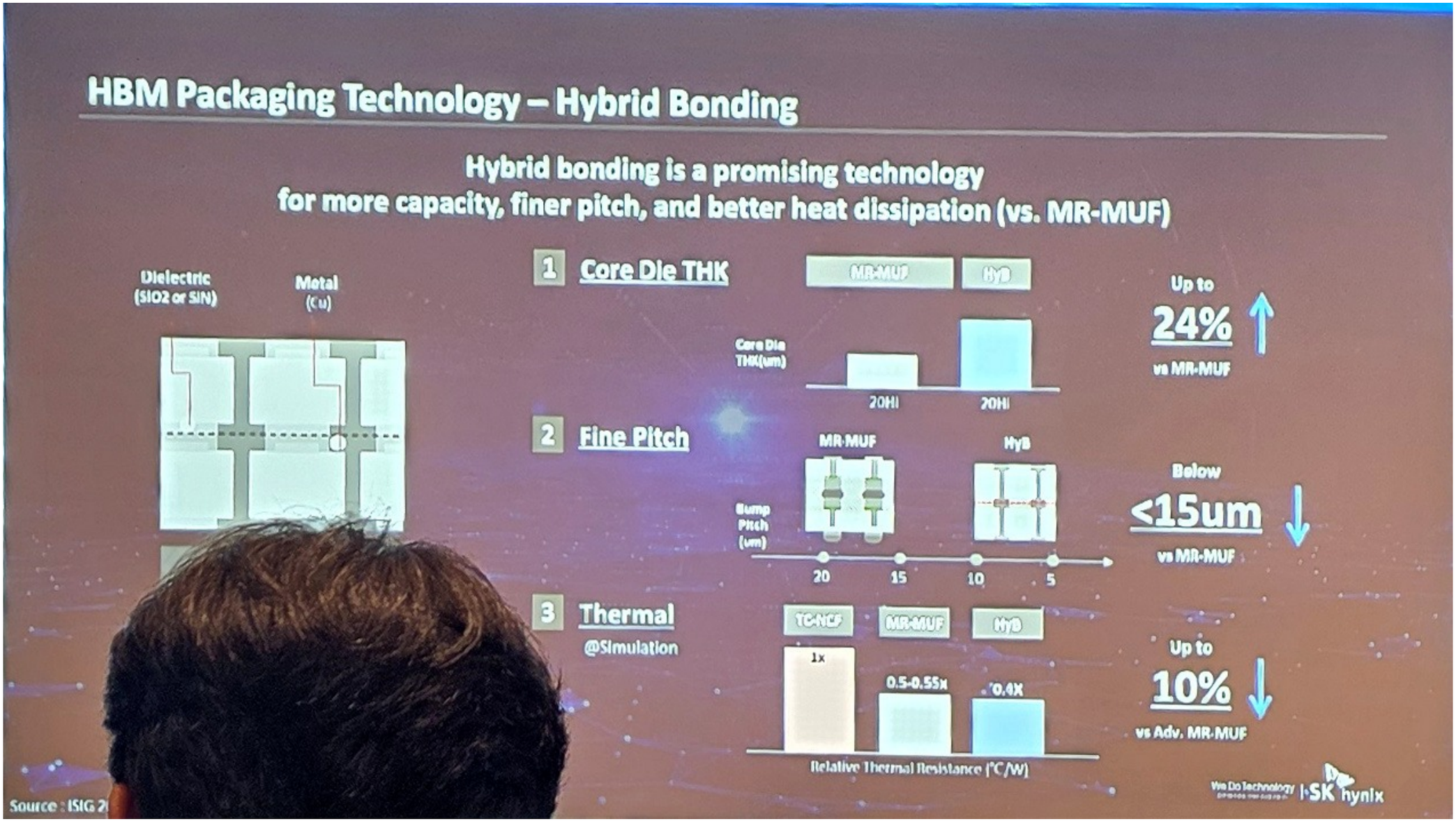
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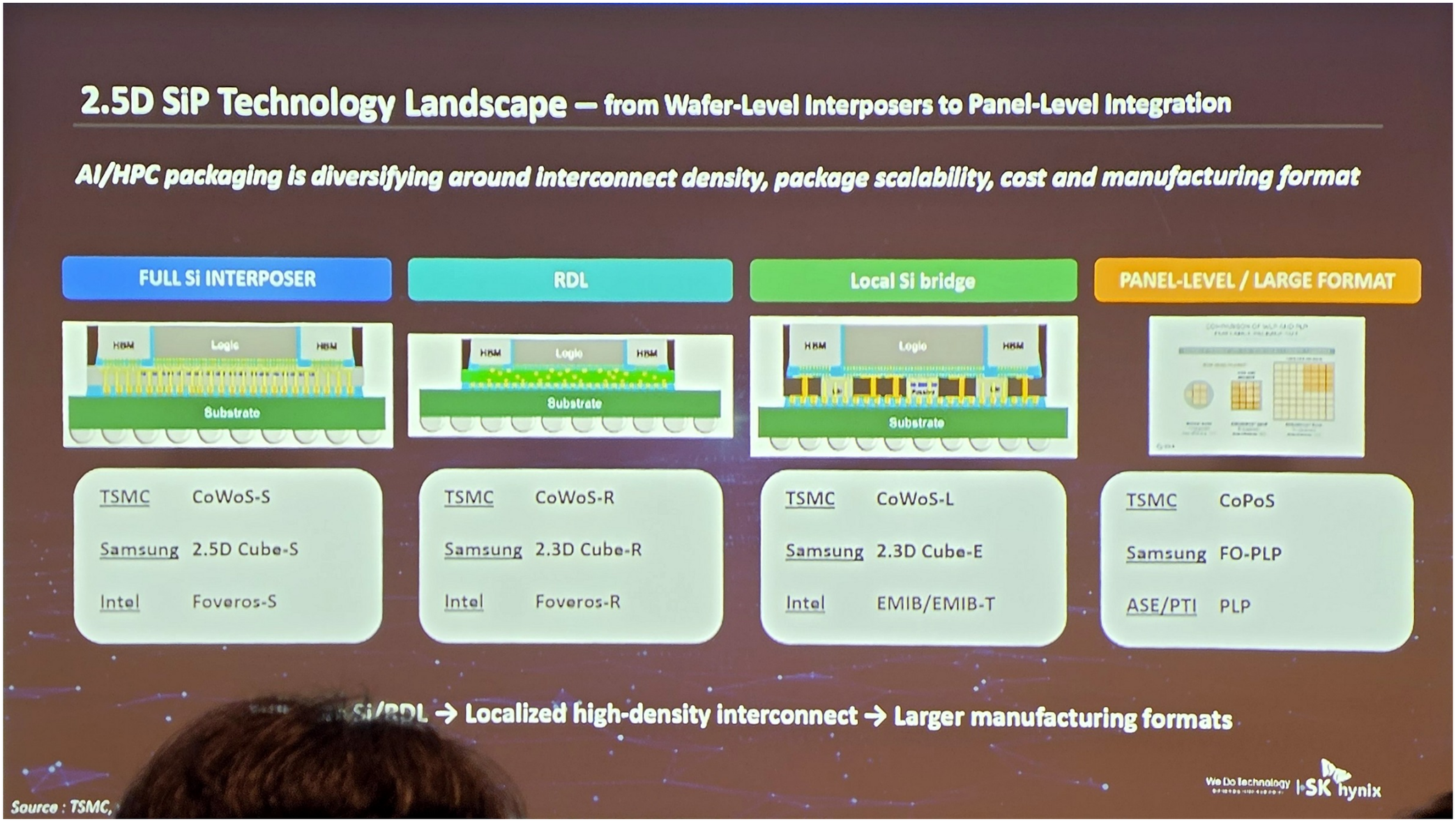
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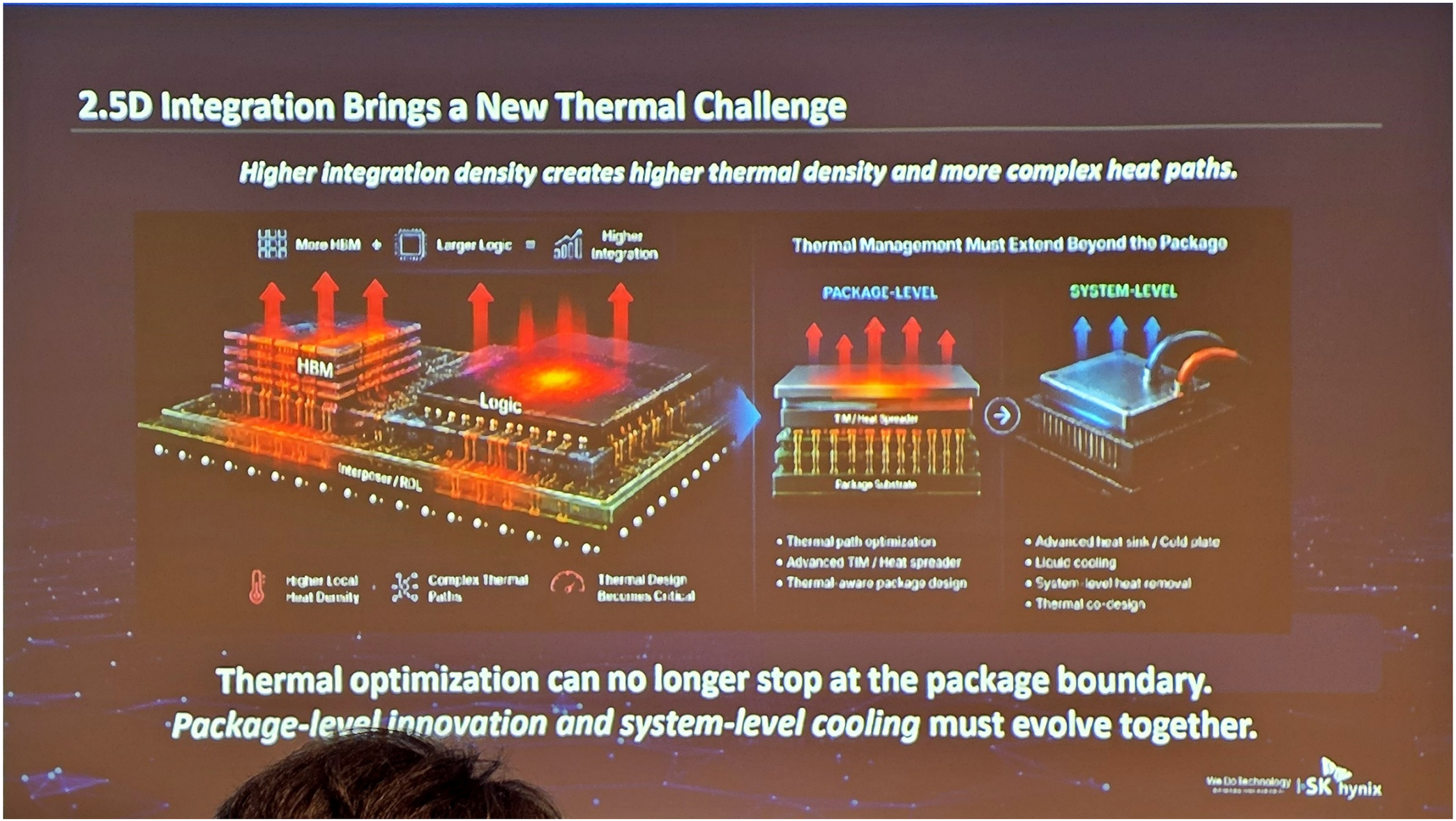
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The Next Leap is to Remove Lateral Distance

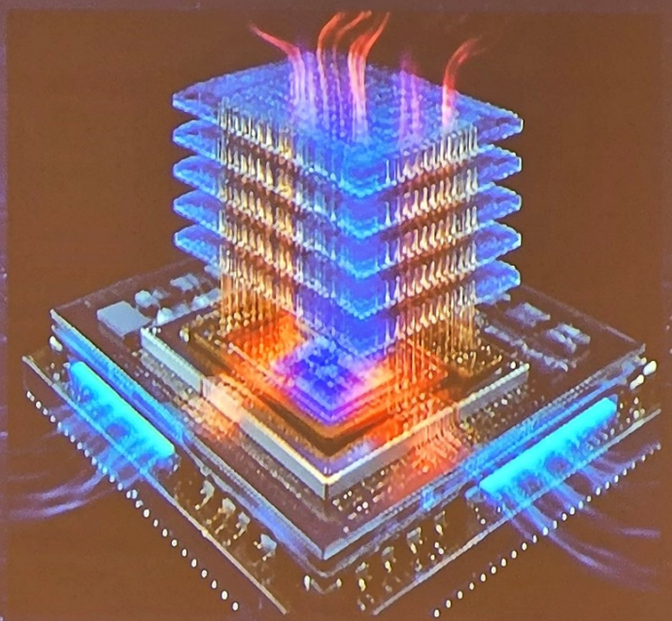
Conventional HBM improves bandwidth by placing stacked DRAM beside the processor.

The future direction is to vertically integrate memory closer to logic.

01 **Shorter data path** Lower I/O energy and latency by reducing physical travel distance.

02 **Higher bandwidth density** Use vertical area rather than only package footprint.

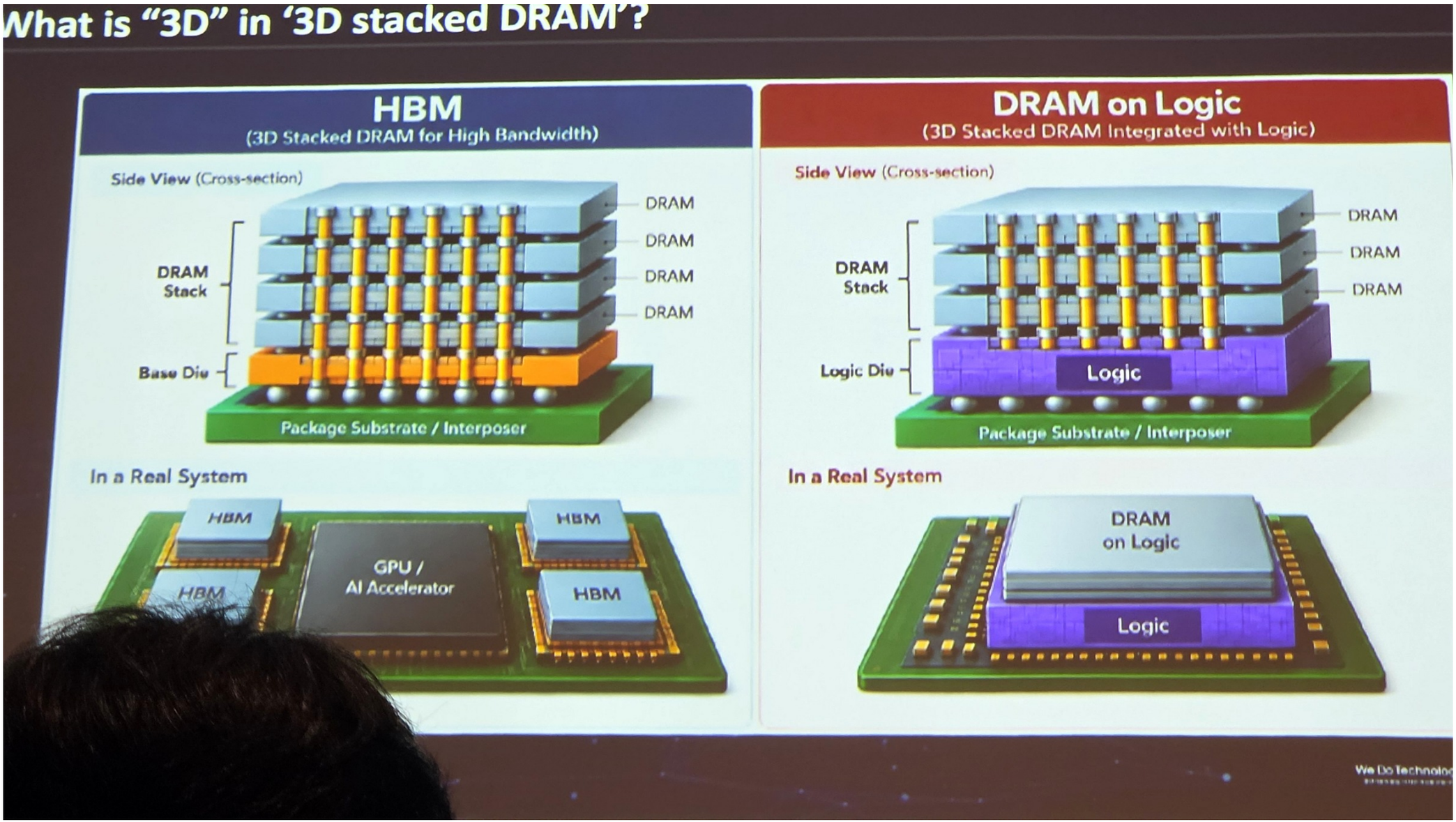
03 **Thermal-aware stack design** Vertical proximity only works if heat has engineered escape paths.



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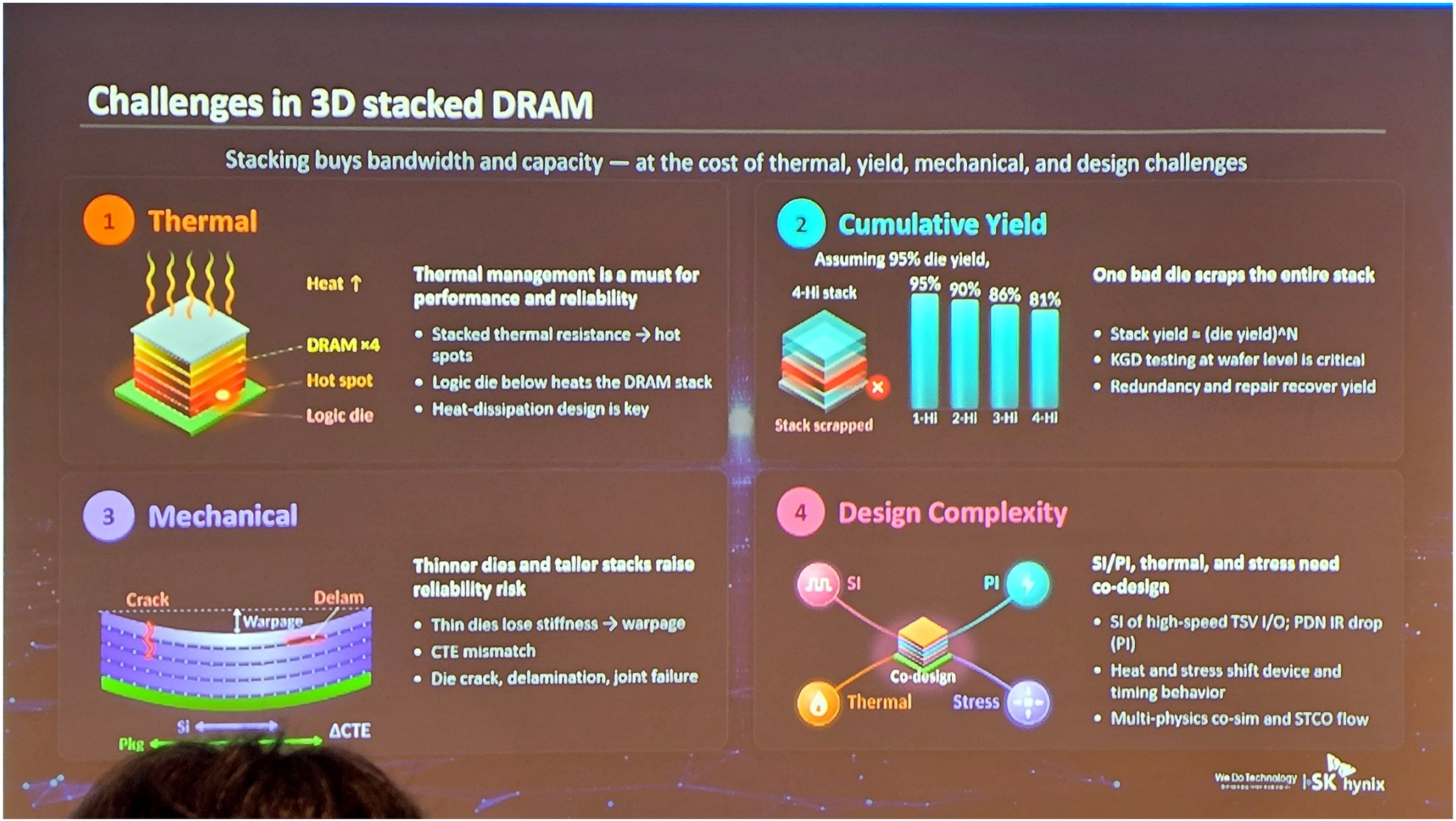
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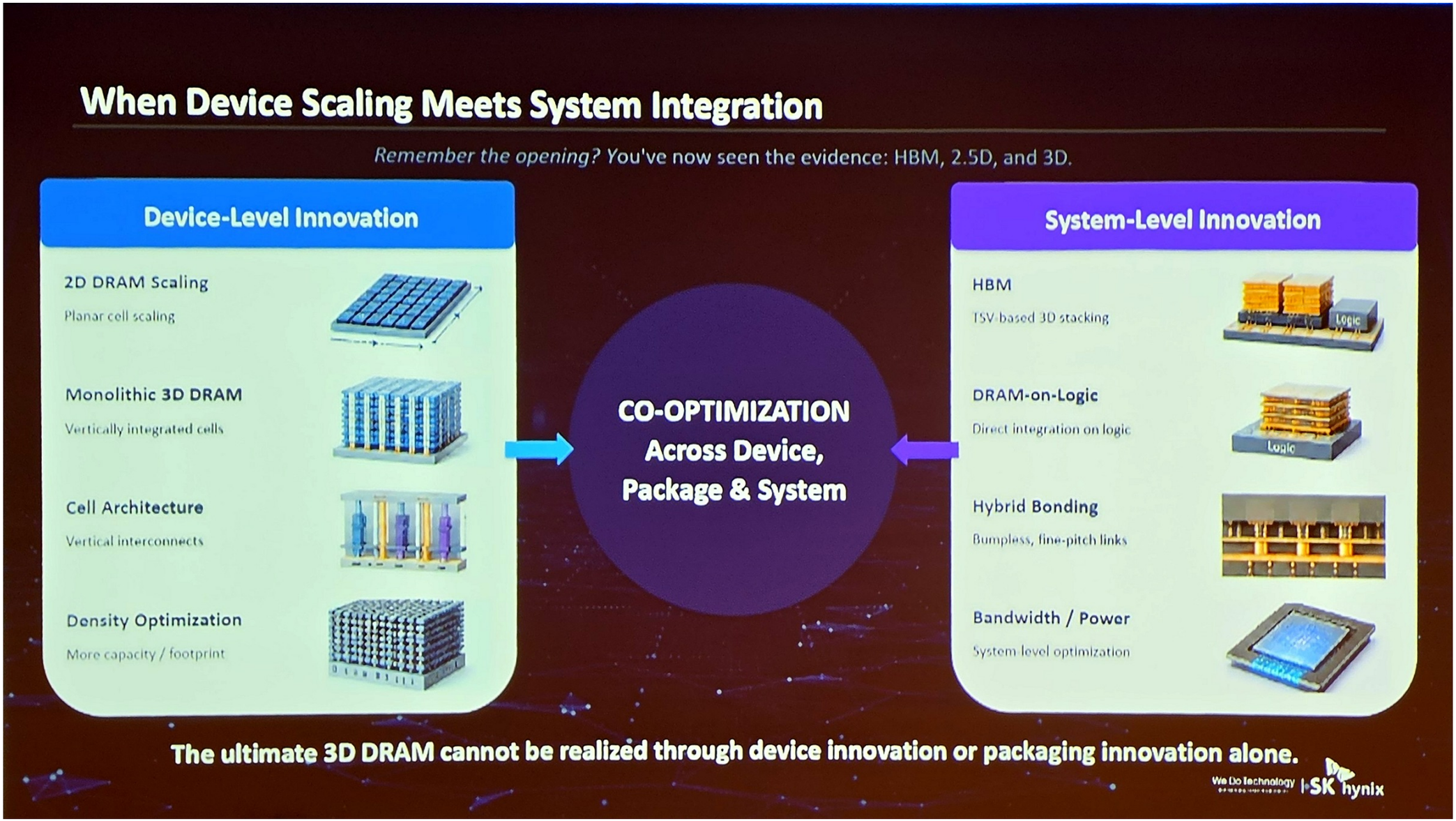
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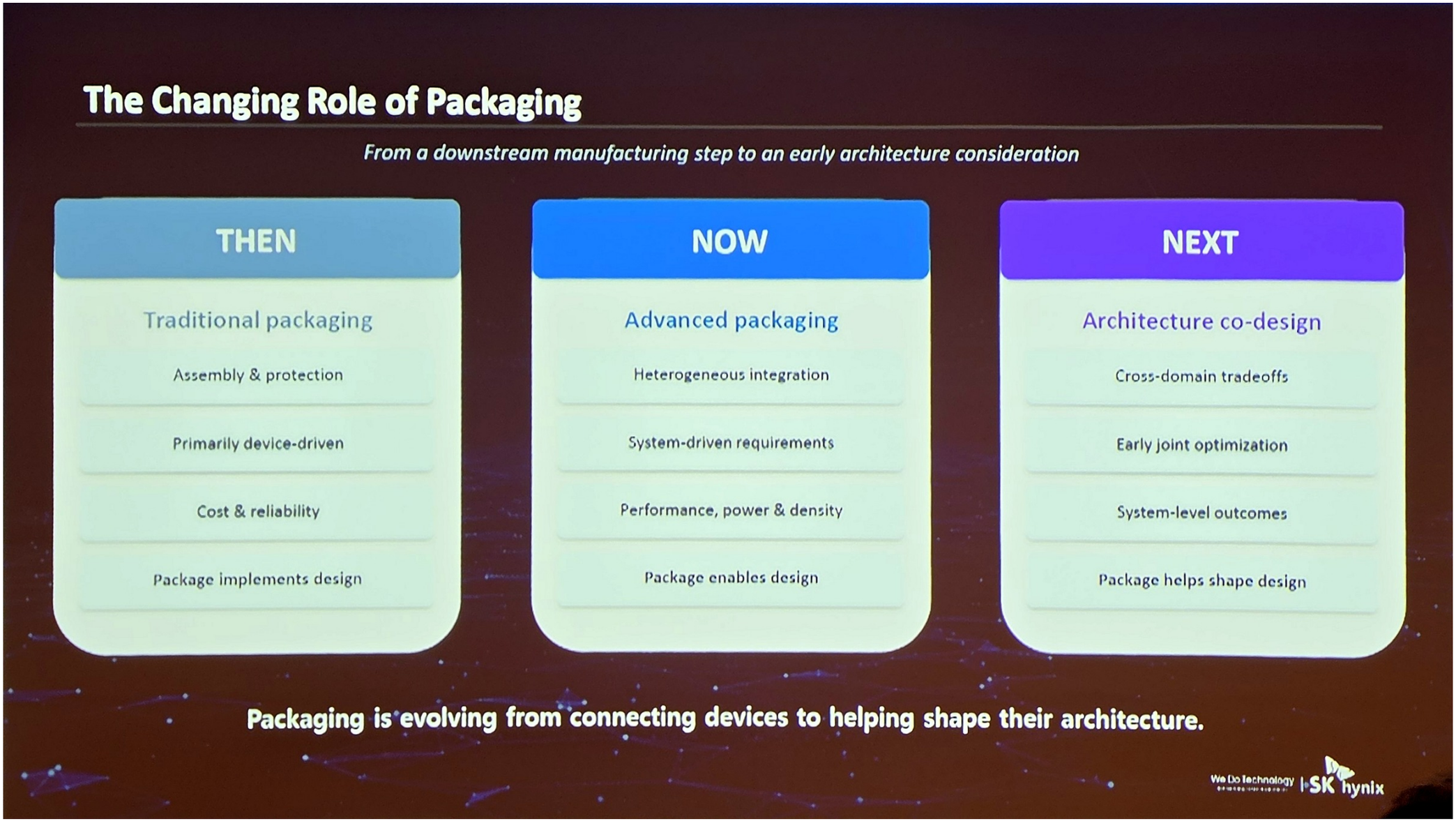
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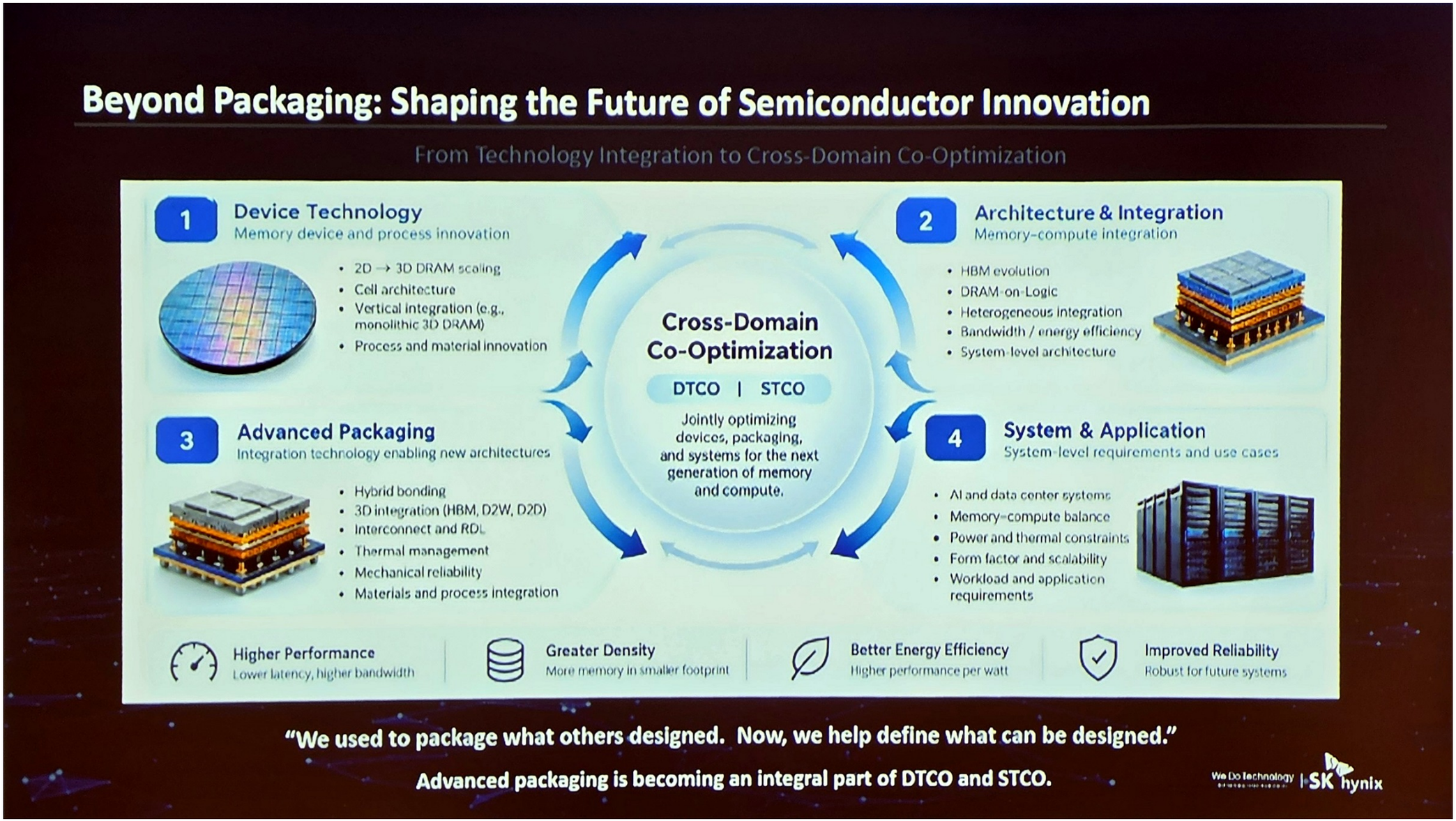
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The Changing Role of Packaging
From a downstream manufacturing step to an early architecture consideration
THEN
Traditional packaging
Assembly & protection
Primarily device-driven
Cost & reliability
Package implements design
NOW
Advanced packaging
Heterogeneous integration
System-driven requirements
Performance, power & density
Package enables design
NEXT
Architecture co-design
Cross-domain tradeoffs
Early joint optimization
System-level outcomes
Package helps shape design
Packaging is evolving from connecting devices to helping shape their architecture.
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